

NBA Shot Prediction

Machine Learning, Biometrics, and the Temporal Dynamics of Basketball

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Motivation

A Data-Rich Environment

Basketball is a sport where every action is meticulously recorded. Understanding the factors that contribute to a successful shot is critical.

Given the fast-paced nature of the game, even marginal improvements in shot prediction can yield massive strategic and analytical advantages.



Related Work & Literature Review

Exploring the Evolution of NBA Shot Prediction Models

Literature Review: Shot Prediction Evolution

Traditional

Tree-Based Baselines

Foundational studies established spatial features (distance, coordinates) as primary predictors.

- **Meehan [1]:** Achieved 68% accuracy using tuned XGBoost on Kaggle data.
- **Oughali et al. [2]:** Confirmed tree-ensembles effectively capture non-linear relationships.
- **Alves et al. [6]:** Identified a stochastic ceiling for accuracy at 64-67%.

Advanced

Tracking & Dynamics

Integration of SportVU data to model defender proximity and intensity.

- **Harmon et al. [3]:** Used multi-agent adversarial trajectories (61% accuracy).
- **Sliz [4]:** Highlighted ball movement and defender distance as critical for 3PT success.
- Focus shifted from static coordinates to dynamic spatial context.

Temporal

Structural Evolution

Analysis of long-term trends and the shifting NBA landscape.

- **Zajac et al. [5]:** Documented the "three-point revolution" across 40 seasons.
- Dynamics are non-stationary; volume and efficiency have shifted drastically.
- **Gap:** This study prioritizes measuring shot distribution over analyzing what factors impact a shots success.

Bridging Research Gaps: Project Innovations

Identified Research Gaps

Biometric Integration

Lack of player physical attributes (height, weight, age) in spatial models.

Temporal Depth

Analysis of *shot affecting features* often ignore potential temporal impacts.

Optimization Rigor

Rare use of hyperparameter-tuned deep neural-networks (Optuna/PyTorch).

Accuracy Ceiling

Limited interpretation of the persistent 64-68% accuracy limit.

Key Innovations & Contributions

Unified Biometric Pipeline

Merging 20 years of shot logs with programmatic nba_api physical data.

Temporal Stability Testing

LightGBM rolling-window analysis across the "3PT Revolution".

Benchmarking Excellence

Comparing XGBoost against Optuna-optimized PyTorch Neural Net architectures.



Project Objectives



Predictive Modeling

Develop and rigorously evaluate machine learning and deep learning models for binary shot outcome prediction (made or missed).



Temporal Analysis

Investigate model stability across eras.



Feature Engineering

Merge shot-by-shot spatial variables with player biometric data to isolate the true physical determinants of shooting success.

Experimental Setup

Implementing the predictive models and temporal analysis.

Data Sourcing & Scope

NBA Shots Datasets

We acquired 20 comprehensive shot-by-shot spatial datasets spanning two decades, from the 2004 through the 2024 NBA seasons. These datasets provided the exact coordinate locations and timing of every shot attempted on the court.

Player Biometric Datasets

Using the *nba_api*, we built a script to create 20 more datasets containing biometric information from every player from every season spanning from 2004-2024

This was an in-depth process that is further outlined in our report and in its own [separate colab file](#).

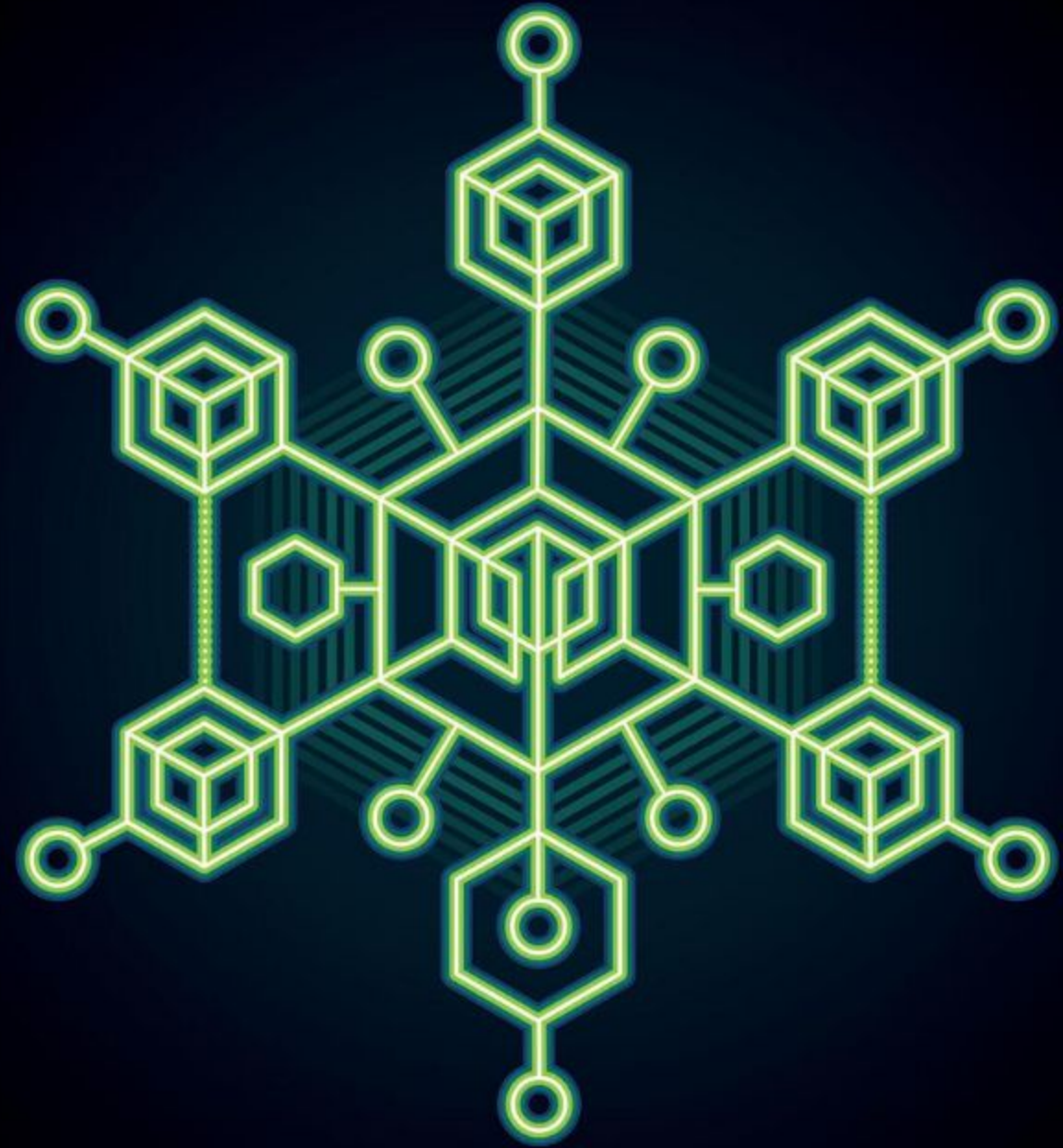
Refining the Dataset

-  **Dataset Consolidation:** Successfully merged disparate spatial shot data with historical player biometrics using common identifiers.
-  **Preventing Data Leakage:** Carefully dropped the EVENT_TYPE column prior to training to ensure models could not peek at the missed/made outcome.
-  **Categorical Pruning:** Removed redundant spatial strings (Zone Names) and categorized rare, low-frequency shooting actions into an "Other" bucket.
-  **Standardization:** Applied One-Hot Encoding for positions and actions, followed by continuous feature scaling for neural network compatibility.

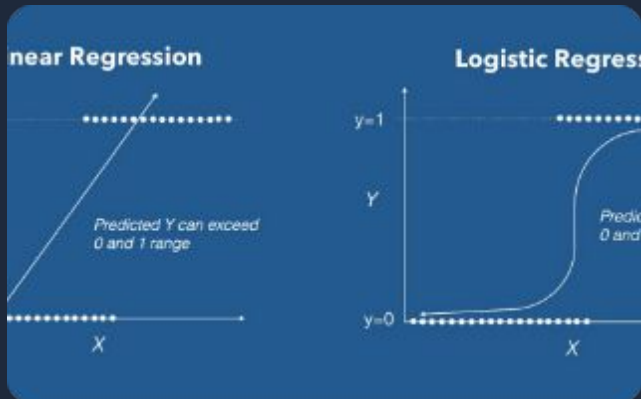
Addressing Multicollinearity

VIF Analysis

Multicollinearity was addressed by calculating Variance Inflation Factor. High VIF scores dictated the removal of all *ACTION_TYPE* columns. This reduced memory footprint and eliminated redundant variables, yielding a precise dataset optimized for machine learning.

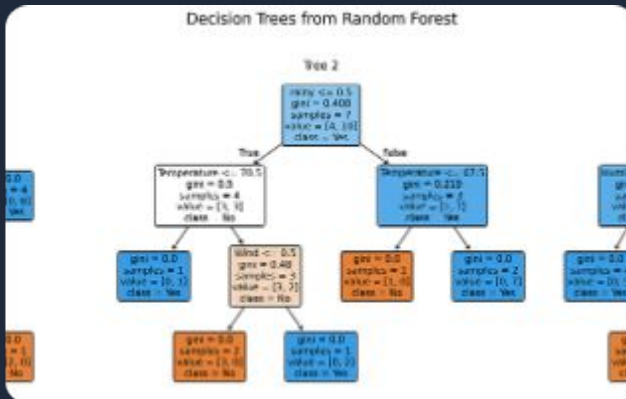


The Machine Learning Pipeline



The Baseline

Logistic Regression configured with balanced class weights to map linear spatial realities.



Ensemble Methods

Random Forest and XGBoost applied to capture complex non-linear variance and interactions.



Deep Learning

Scikit-learn MLPs and heavily tuned PyTorch Neural Networks for optimal performance.

Model Configuration & Tuning



Baseline & Ensembles

Logistic Regression served as the baseline with tuned regularization. Random Forest and XGBoost architectures were rigorously optimized for depth, estimators, and learning rates via RandomizedSearchCV.



Deep Learning

Custom PyTorch feedforward neural networks were deployed utilizing mini-batch gradient descent. The Adam optimizer was paired with strict early stopping protocols to prevent validation overfitting.



Hyperparameter Search

The Optuna framework was leveraged to systematically explore complex parameter spaces. Multiple Bayesian trials honed in on optimal learning rates, dropout percentages, and hidden layer dimensions.

Temporal Analysis Strategy

Rolling Windows

LightGBM models were trained on fixed-size historical periods that continuously moved forward in time. This strategy evaluates how rapidly models can adapt to evolving NBA gameplay and whether discarding older data improves short-term predictive accuracy.

Expanding Windows

Models were trained on incrementally growing datasets, retaining all prior history. Tested strictly against a fixed modern baseline, this approach isolated the intrinsic value of deep historical context versus the pitfalls of recency bias.

Head to Head

Models were trained on the oldest 30% of data and the newest 30% of data. These are tested against each other on test set to see if newer data is more relevant than the older data for training our models.

Results and Discussion

Analyzing the performance ceilings and limitations across the
predictive architectures.

Establishing the Baseline

LOGISTIC REGRESSION

59.0%

Test Accuracy

Macro F1-Score: 0.59

CONFUSION MATRIX (N=42,311)

14,143

True Negatives

8,803

False Positives

8,545

False Negatives

10,820

True Positives

Key Takeaways & Setup

- **Predictive Signal Exists**
Spatial coordinates and biometrics provide a tangible edge over random chance, confirming feature validity.
- **The Stochastic Reality**
59% accuracy highlights immense noise; unrecorded micro-physical events heavily influence outcomes.
- **Linear Limitations**
Linear decision boundaries plateau quickly; a purely linear combination of spatial and biometric data cannot capture the dynamic nature of an NBA game.
- **Configuration Details**
Balanced class weights applied to mitigate imbalance (miss vs make) and establish a floor for linear assumptions; proved that the feature engineering and preprocessing worked.

Ensemble & MLP Comparison

Random Forest (Optimized)

BEST F1

0.511

Validation F1

0.507

Test F1

60.9%

Test Accuracy

Neural Network (MLP)

BEST ACCURACY

0.509

Validation F1

0.511

Test F1

61.4%

Test Accuracy

XGBoost

0.499

Val F1

0.498

Test F1

60.6%

Test Accuracy

Mapping Non-Linear Interactions

Key Findings

- **Capturing Non-Linearity:** Upgrading to architectures that map complex variables, such as player height interacting with floor positioning.
- **The "Hard Ceiling" of Noise:** Advanced models clustered at 60–61% accuracy, revealing signal limits without defensive proximity context.
- **Performance Clustering:** Tight grouping suggests mathematical exhaustion of the current spatial/biometric feature set.

RANDOM FOREST

BEST F1

0.511

Validation F1-Score

CONFUSION MATRIX

17,226
True Negatives

5,721
False Positives

10,839
False Negatives

8,526
True Positives

NEURAL NET (MLP)

BEST ACCURACY

61.4%

Test Accuracy

XGBOOST

0.499

Validation F1-Score

Lagged in specific tabular configuration

PyTorch "ShotNet" Optimization

Custom PyTorch Architecture

Built a custom Feed-Forward network to exert granular control after ensemble models hit a performance ceiling.

Class Imbalance Handling

Used *BCEWithLogitsLoss* with a dynamic positive class weight to force the network to prioritize correctly predicting "made" shots.

62.2%

Peak Accuracy

Spatial Regularization

Injected 1D Batch Normalization and Dropout rates up to 50% to prevent the model from memorizing exact coordinates.

Optuna Bayesian Search

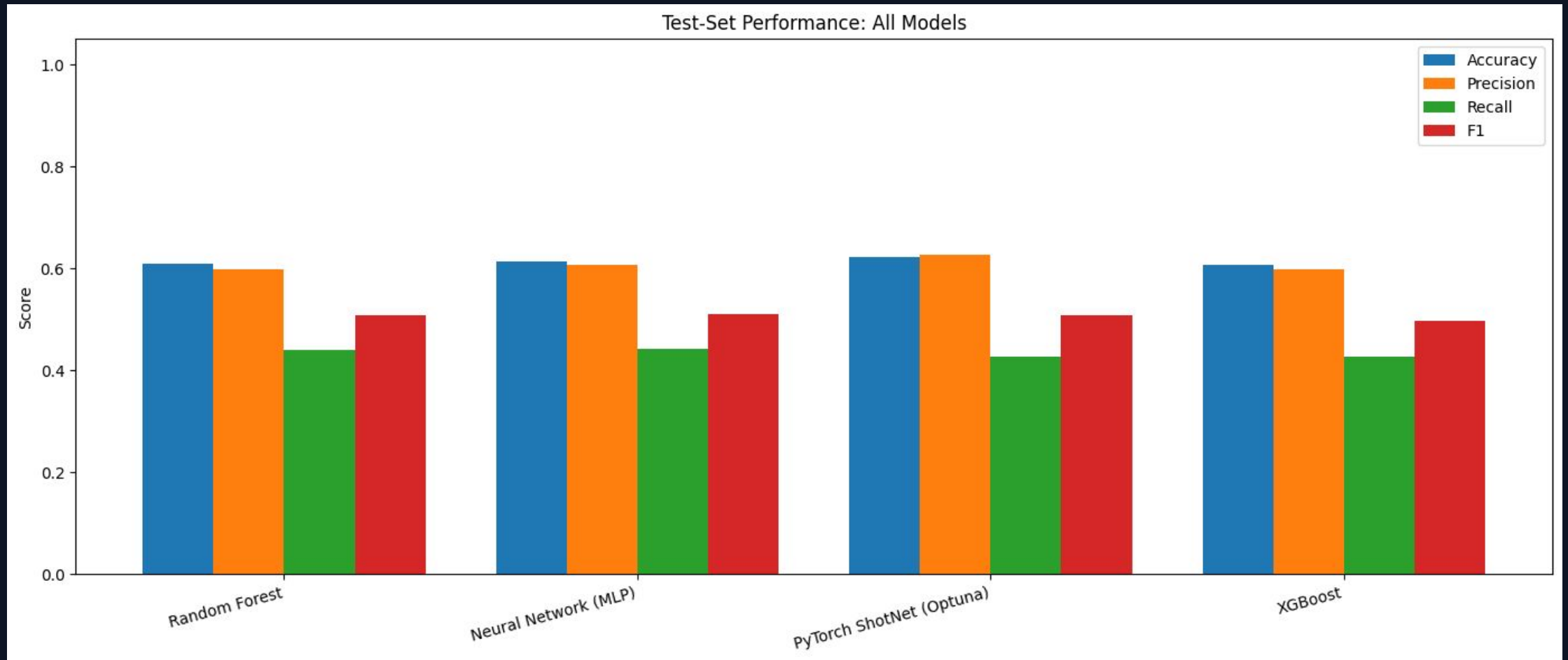
Leveraged TPE sampling across 30 trials to learn from previous results and hone in on the ideal learning rate and layer depth.

Optimization Objective:
Maximize Validation F1-Score

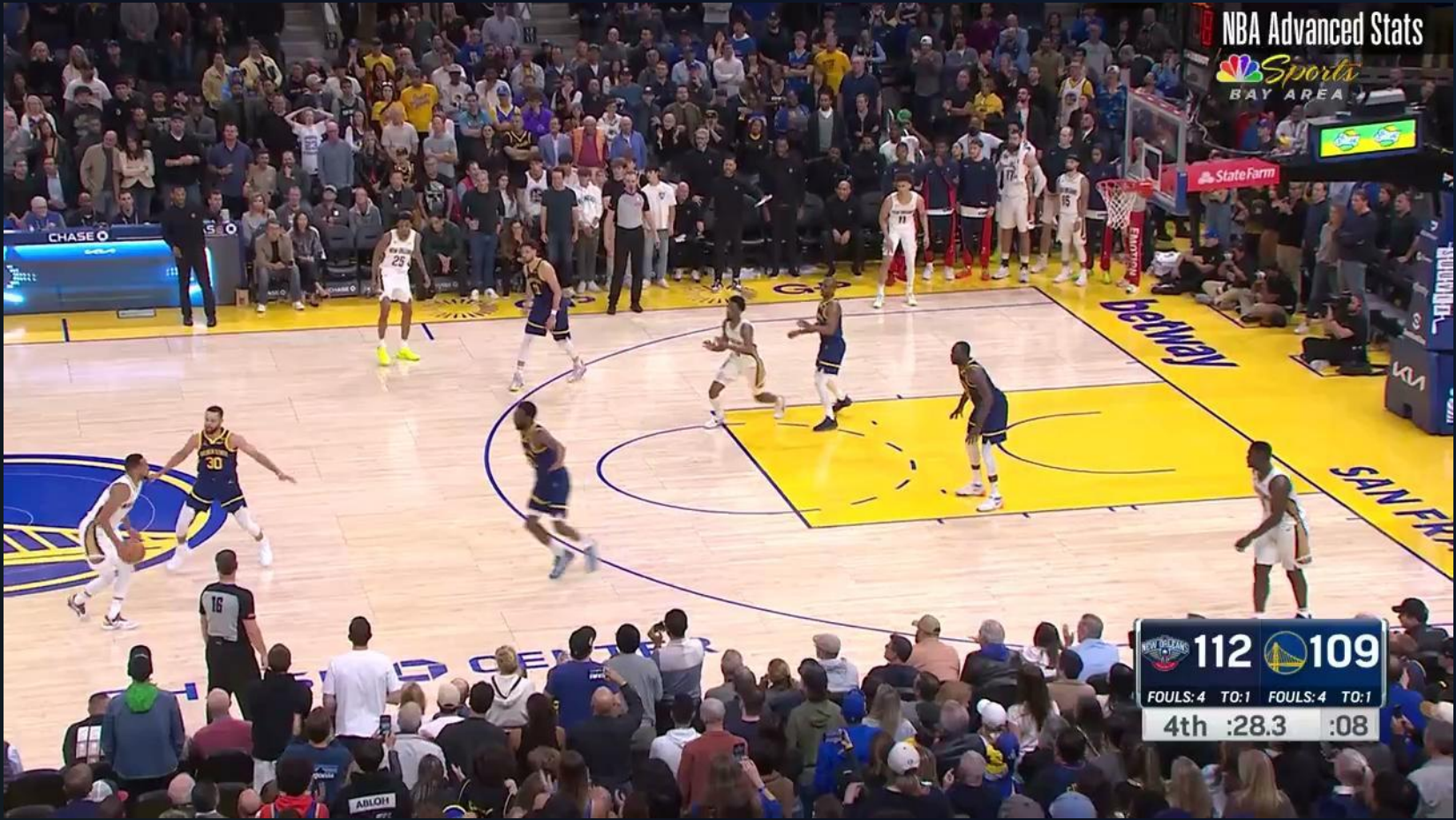
The network failed to identify **57.2%** of all successful shots (missing 11,083 out of 19,365 actual test-set makes).

Despite reaching an overall accuracy of 62.2%, the PyTorch ShotNet struggled significantly with positive recall. It correctly flagged over 18,000 missed shots but incorrectly predicted a miss for over 11,000 shots that players actually made.

Test-Set Performance



Incorrect Example: Predicted Make, Actual Miss



PLAYER_ID	PLAYER_NAME	GAME_ID	QUARTER	MINS_LEFT	SECS_LEFT	Actual	Predicted	LOC_X	LOC_Y	SHOT_DISTANCE	AGE	HEIGHT	WEIGHT
1629627	Zion Williamson	22301182	4	0	21	0	1	-2.7	10.45	5	26	79"	195"

Correct Example: Predicted *Make*, Actual *Make*



PLAYER_I D	PLAYER_N AME	GAME_ID	QUARTER	MINS_LEF T	SECS_LEF T	Actual	Predicted	LOC_X	LOC_Y	SHOT_DISTANCE	AGE	HEIGHT	WEIGHT	POSITION
2974128	Alperen Sengun	22300216	4	0	17	1	1	0	4.65	0	28	83"	245	C

62.2%

Predictive Accuracy Ceiling

The 62.2% threshold represents a persistent "stochastic ceiling" in NBA shot prediction. While model complexity has increased, the inherent variability of human performance remains a primary constraint.

This performance limit is largely dictated by the lack of defensive tracking data and the extreme outliers of elite shooting skill that defy standard spatial modeling.

The Precision-Recall Tradeoff

11,083

False Negatives Generated

Missing the "Made" Shots

The network failed to identify **57.2%** of all successful shots (missing 11,083 out of 19,365 actual test-set makes).

Despite reaching an overall accuracy of 62.2%, the PyTorch ShotNet struggled significantly with positive recall. It correctly flagged over 18,000 missed shots but incorrectly predicted a miss for over 11,000 shots that players actually made.

The Precision-Recall Anomaly



Missing Latent Variables: The dataset completely lacks high-fidelity defensive context. Without knowing defender proximity and contest intensity, models are forced to regress toward historical spatial means.



Unquantifiable Human Skill: Top-tier NBA athletes frequently convert attempts characterized by extreme mathematical improbability, defying standard classification boundaries.

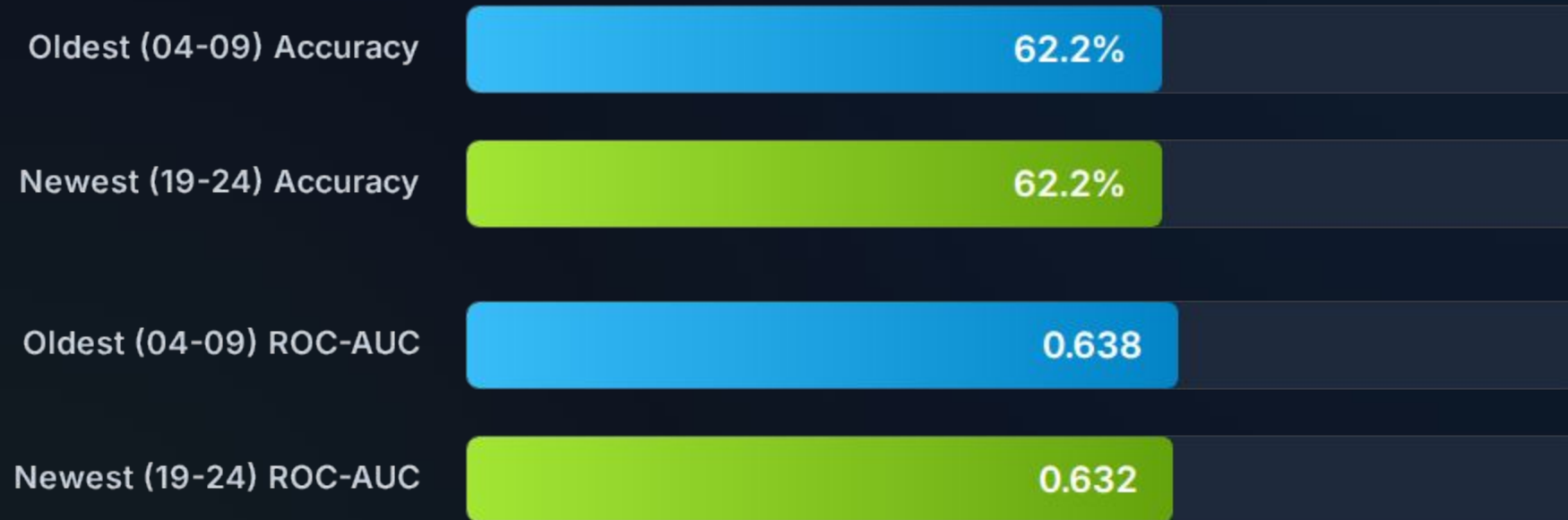


Algorithm Conservatism: Due to average jump-shot conversion rates fluctuating between 35-45%, binary classifiers optimize for global accuracy by defaulting to a negative prediction, sacrificing positive recall.

Temporal Analysis & Recency Bias

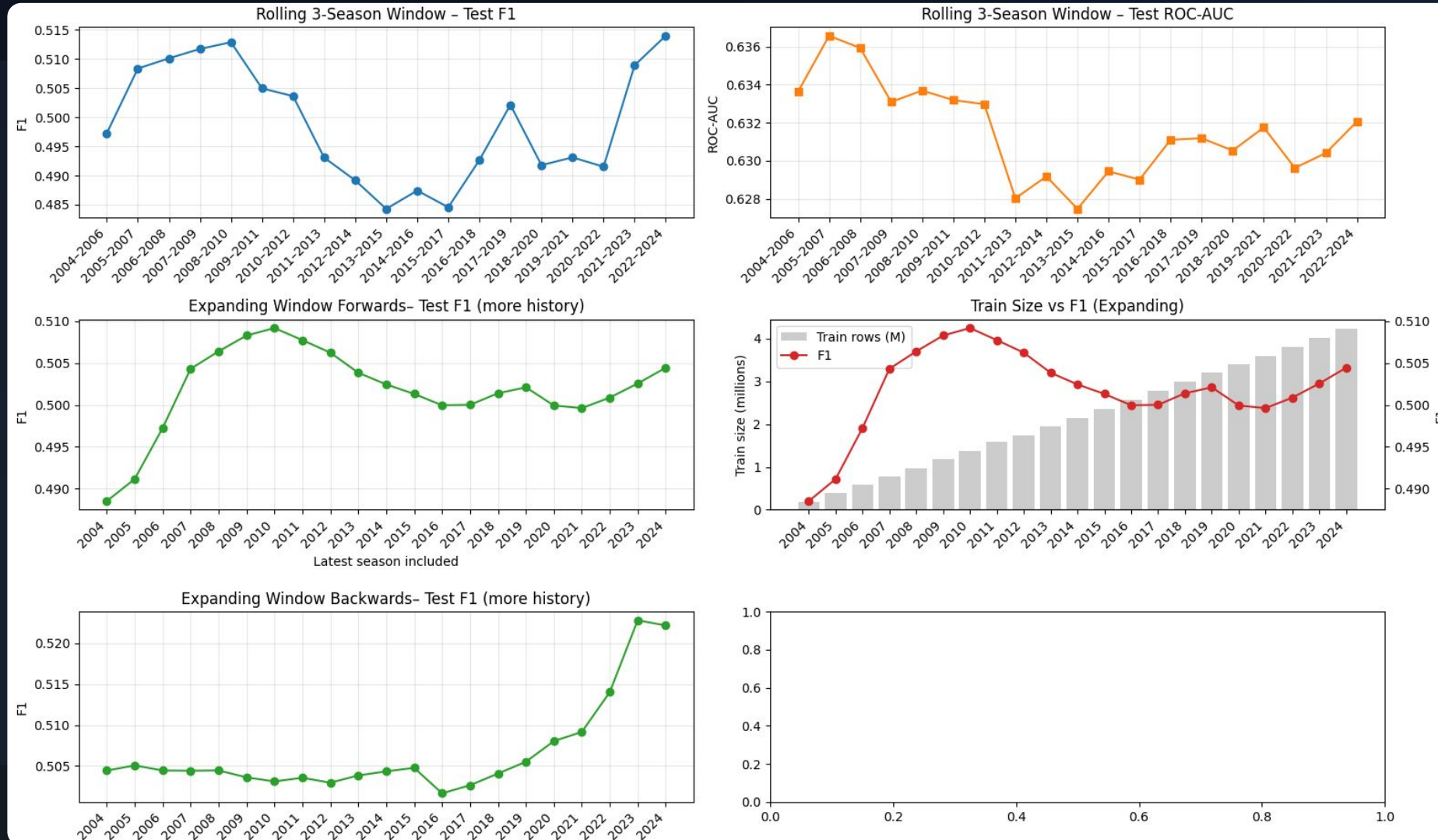
Testing the assumption that older data degrades modern predictive power in the era of the "Three-Point Revolution."

Oldest 30% vs. Newest 30% (Head-to-Head)



Models trained exclusively on early-2000s data predicted modern test outcomes with virtually identical accuracy to models trained purely on modern data.

Temporal Influence on Shot-Prediction Performance



Visual analysis of model stability and accuracy trends across rolling and expanding historical windows (2004–2024).

Temporal Stability & Recency Bias Results

Rolling 3-Season Window Stability

When evaluating rolling 3-season windows, Test AUC remained remarkably stable between 0.628 and 0.636, showing no severe degradation regardless of the era. Model performance maintained a Test AUC of approximately 0.630 across the two-decade span, challenging the initial hypothesis regarding recency bias.

Expanding Window Resilience

Progressively adding historical data (2004–2024) demonstrated a slow but consistent improvement in model performance. Training exclusively on the 2004 season yielded an AUC of 0.630, while including all seasons up to 2024 elevated the AUC to 0.640. This confirms that more data reinforces universal shooting fundamentals over outdated stylistic trends.

Head-to-Head Comparison

A model trained on the oldest 30% of data (2004–2009) predicted modern outcomes with ~62.2% accuracy and 0.638 AUC, virtually identical to a model trained on the newest 30% (2019–2024) which recorded 0.632 AUC. Shifts in strategy have not materially affected shooting prediction.



Cumulative Window AUC Peak

Key Conclusions



The 62% Ceiling

Pure spatial and biometric modeling hits a definitive performance limit. Algorithm complexity cannot out-compute missing game context.



Volume vs. Recency

Expanding the training window to include all historical data improved the AUC to 0.640. More data reinforces core fundamentals better than restricted modern subsets.



Timeless Mechanics

While shot volume and strategy have radically changed, the fundamental physical laws of a basketball entering a hoop remain entirely unchanged over 20 years.

References

[1] B. Meehan, "Predicting NBA shots," Stanford University CS229 Machine Learning Project, 2017.

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[3] M. Harmon et al., "Predicting shot making in basketball learnt from adversarial multiagent trajectories," arXiv, 2016.

[4] B. A. Sliz, "An investigation of three-point shooting through an analysis of NBA player tracking data," arXiv, 2017.

[5] T. Zając et al., "Long-term trends in shooting performance in the NBA: An analysis of 40 consecutive seasons," IJERPH, 2023.

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[7] maxwell, "NBA Shots," Kaggle, 2024. Available: kaggle.com/datasets/maxwell/nba-shots

[8] J. Fodera, "Biometric Attributes of NBA Players per Season," Kaggle, 2026. Available: kaggle.com/datasets/joefodera/biometric-attributes

Questions?

Thank you for your time and attention.

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